

Time Dispersion and Frequency Selectivity

Derivations and Applied Case Study at 60 GHz - ELEC-H415 (Chapter 1)

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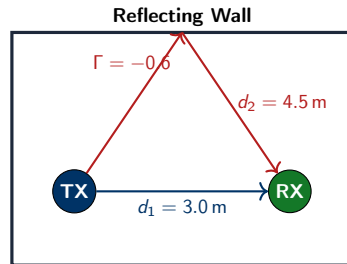
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Applied Case Study: Indoor 60 GHz Wireless Communication

Indoor Measurement Setup (Course Fig. 1.2–1.3):

- Carrier frequency: $f_c = 60$ GHz ($\lambda = c/f_c = 5$ mm).
- Baseband bandwidth: $B = 2.0$ GHz ($f \in [-1, +1]$ GHz).
- Channel geometry:
 - Direct Path (Path 1):** Distance $d_1 = 3.0$ m.
 - Wall Reflected Path (Path 2):** Distance $d_2 = 4.5$ m, reflection coefficient $\Gamma = -0.6 = 0.6e^{j\pi}$.



Objective

Derive how multipath delays distort the transmitted spectrum $|H(f)|$ across the 2 GHz bandwidth from physical principles.

Step 1A: Passband Propagation for a Single Path

Let the transmitter emit an unmodulated carrier at frequency f_c :

$$x_{\text{RF}}(t) = \cos(2\pi f_c t)$$

Along physical path n , the wave propagates over distance d_n with speed c :

- **Propagation Delay:** $\tau_n = d_n/c$.
- **Channel Attenuation:** $a_n \in \mathbb{R}^+$
- **Phase Shift:** $\phi_n \in [0, 2\pi)$

Received Passband Signal on Path n

$$\begin{aligned} y_{\text{RF},n}(t) &= a_n \cos(2\pi f_c(t - \tau_n) + \phi_n) \\ &= a_n \cos(2\pi f_c t + (\phi_n - 2\pi f_c \tau_n)) \end{aligned}$$

The carrier undergoes a total phase rotation: $\theta_n = \phi_n - 2\pi f_c \tau_n$.

Step 1B: Passband to Baseband

Using complex analytic signal representation:

$$y_{\text{RF},n}(t) = \text{Re} \{ y_n(t) e^{j2\pi f_c t} \}$$

where $y_n(t) \in \mathbb{C}$ is the baseband representation of the received signal.

$$\begin{aligned} y_{\text{RF},n}(t) &= a_n \cos(2\pi f_c t + (\phi_n - 2\pi f_c \tau_n)) \\ &= \text{Re} \{ (a_n e^{j\phi_n} e^{-j2\pi f_c \tau_n}) e^{j2\pi f_c t} \} \end{aligned}$$

Complex Channel Amplitude α_n

$$y_n(t) = a_n e^{j\phi_n} e^{-j2\pi f_c \tau_n} \triangleq \alpha_n$$

When the carrier is modulated by an arbitrary baseband signal $x(t)$:

$$x_{\text{RF}}(t) = \text{Re} \{ x(t) e^{j2\pi f_c t} \} \implies y_n(t) = \alpha_n x(t - \tau_n)$$

Step 1C: Phase Sensitivity at 60 GHz vs. Sub-6 GHz

The carrier phase term is directly proportional to physical distance d_n :

$$\theta_n = -2\pi f_c \tau_n = -2\pi f_c \frac{d_n}{c} = -2\pi \frac{d_n}{\lambda_c}$$

Taking the spatial derivative with respect to path length d_n :

$$\boxed{\frac{\partial \theta_n}{\partial d_n} = -\frac{2\pi}{\lambda_c} = -\frac{2\pi f_c}{c}}$$

Sub-6 GHz ($f_c = 1$ GHz)

- $\lambda_c = 30$ cm.
- $\Delta d = 15$ cm $\implies \Delta\theta = \pi$ (inversion).
- Large movements needed to alter α_n .

5G mmWave ($f_c = 60$ GHz)

- $\lambda_c = 5$ mm.
- $\Delta d = 2.5$ mm $\implies \Delta\theta = \pi$ (inversion).
- **Small movements** completely randomize the complex amplitude phase.

Step 2: Multipath Superposition & Impulse Response $h(\tau)$

In a real propagation channel, the received signal is the sum of N distinct physical paths:

$$y(t) = \sum_{n=1}^N a_n e^{j\phi_n} e^{-j2\pi f_c \tau_n} x(t - \tau_n) = \sum_{n=1}^N \alpha_n x(t - \tau_n)$$

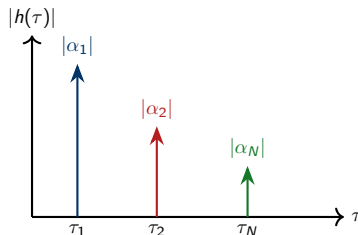
Dirac Excitation $x(t) = \delta(t)$:

$$y(t) = \sum_{n=1}^N \alpha_n \delta(t - \tau_n)$$

Baseband Channel Impulse Response

$$h(\tau) = \sum_{n=1}^N \alpha_n \delta(\tau - \tau_n)$$

where $\alpha_n = a_n e^{j\phi_n} e^{-j2\pi f_c \tau_n}$.



- $h(\tau)$ is a sum of Dirac distributions.
- Fully characterizes channel physics in delay domain τ .

Step 3: Input-Output Relationship as a Convolution

Using the "sifting" property of Dirac distributions:

$$x(t - \tau_n) = \int_0^\infty \delta(\tau - \tau_n) x(t - \tau) d\tau$$

Substituting into $y(t) = \sum_{n=1}^N \alpha_n x(t - \tau_n)$:

$$y(t) = \sum_{n=1}^N \alpha_n \int_0^\infty \delta(\tau - \tau_n) x(t - \tau) d\tau = \int_0^\infty \underbrace{\left[\sum_{n=1}^N \alpha_n \delta(\tau - \tau_n) \right]}_{=h(\tau)} x(t - \tau) d\tau$$

Convolution Product

$$y(t) = \int_0^\infty h(\tau) x(t - \tau) d\tau = (h * x)(t)$$

Physical Interpretation of Time Dispersion: When the transmitter sends one single impulse, the receiver perceives multiple delayed copies spread over time.

Step 4: Derivation of the Channel Transfer Function $H(f)$

We compute the Fourier transform of $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau$ with respect to time t :

$$Y(f) = \int_{-\infty}^{\infty} y(t)e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau \right] e^{-j2\pi ft} dt$$

Interchanging integrals via Fubini's theorem:

$$Y(f) = \int_{-\infty}^{\infty} h(\tau) \left[\int_{-\infty}^{\infty} x(t - \tau)e^{-j2\pi ft} dt \right] d\tau$$

Applying substitution $u = t - \tau \Rightarrow t = u + \tau$ and $dt = du$:

$$\begin{aligned} Y(f) &= \int_{-\infty}^{\infty} h(\tau) \left[\int_{-\infty}^{\infty} x(u)e^{-j2\pi f(u+\tau)} du \right] d\tau \\ &= \left(\int_{-\infty}^{\infty} h(\tau)e^{-j2\pi f\tau} d\tau \right) \underbrace{\left(\int_{-\infty}^{\infty} x(u)e^{-j2\pi fu} du \right)}_{=X(f)} \end{aligned}$$

Baseband Transfer Function

$$H(f) \triangleq \int_{-\infty}^{\infty} h(\tau)e^{-j2\pi f\tau} d\tau \Rightarrow Y(f) = H(f)X(f)$$

Step 5: Transfer Function for Discrete Multipath Components

Injecting $h(\tau) = \sum_{n=1}^N \alpha_n \delta(\tau - \tau_n)$ into the Fourier transform definition:

$$\begin{aligned} H(f) &= \int_{-\infty}^{\infty} \left[\sum_{n=1}^N \alpha_n \delta(\tau - \tau_n) \right] e^{-j2\pi f \tau} d\tau \\ &= \sum_{n=1}^N \alpha_n \int_{-\infty}^{\infty} \delta(\tau - \tau_n) e^{-j2\pi f \tau} d\tau \end{aligned}$$

Applying the Dirac integral:

$$H(f) = \sum_{n=1}^N \alpha_n e^{-j2\pi f \tau_n} = \sum_{n=1}^N a_n e^{j\phi_n} e^{-j2\pi f_c \tau_n} e^{-j2\pi f \tau_n}$$

Total Path Phase

$$H(f) = \sum_{n=1}^N a_n e^{j\phi_n} e^{-j2\pi(f_c + f)\tau_n}$$

- f_c sets the microscopic path phase (60 GHz carrier).
- f sets the frequency-dependent phase slope across the baseband spectrum (± 1 GHz).

Step 6: Mathematical Proof of Frequency Selectivity

Case 1: No Time Dispersion ($\tau_n = \tau_1, \forall n$)

$$H(f) = e^{-j2\pi f \tau_1} \sum_{n=1}^N \alpha_n$$

Computing the magnitude:

$$|H(f)| = \left| e^{-j2\pi f \tau_1} \right| \cdot \left| \sum_{n=1}^N \alpha_n \right| = \left| \sum_{n=1}^N \alpha_n \right|$$

- $|H(f)|$ is **constant over bandwidth B** .
- Signal spectrum is not distorted.
- **Non-frequency-selective (Flat channel)**.

Case 2: Arbitrary Delays ($\exists \tau_n \neq \tau_m$)

$$|H(f)| = \left| \sum_{n=1}^N \alpha_n e^{-j2\pi f \tau_n} \right|$$

- $|H(f)|$ **varies with baseband frequency f** .
- Multipaths interfere constructively at some frequencies and destructively at others.
- **Frequency-selective channel**.

Fundamental Principle

Time dispersion during propagation $\Longleftrightarrow$ Frequency selectivity across bandwidth.

Step 7: The Physical Interference Process

Every copy n arrives with baseband phase $-2\pi f\tau_n$.

Phase Difference Between Two Paths n and m

$$\Delta\Phi_{nm}(f) = -2\pi f(\tau_n - \tau_m)$$

- **Frequency Dependence:**

- When $\Delta\Phi_{nm}(f) = 2k\pi$: constructive interference $\implies |H(f)|$ reaches a local maximum.
- When $\Delta\Phi_{nm}(f) = (2k + 1)\pi$: destructive interference $\implies |H(f)|$ reaches a local minimum.

- **Delay Difference Dependence:** The speed of phase rotation with respect to f is:

$$\left| \frac{d\Delta\Phi_{nm}}{df} \right| = 2\pi|\tau_n - \tau_m|$$

- **Physical Consequence:** The larger the delay spread $|\tau_n - \tau_m|$, the faster the phase changes with f , making **frequency selectivity more severe**.

Step 8A: 60 GHz Case Study, Exact Path Calculation

1. Direct Path ($n = 1$):

- Delay:

$$\tau_1 = \frac{d_1}{c} = \frac{3.0 \text{ m}}{3 \times 10^8 \text{ m/s}} = 10.0 \text{ ns}$$

- Free-space normalization: $a_1 = 1.0$, $\phi_1 = 0$.
- Complex gain:

$$\begin{aligned}\alpha_1 &= 1.0 \cdot e^{-j2\pi(60 \times 10^9)(10 \times 10^{-9})} \\ &= 1.0 \cdot e^{-j1200\pi} = 1.0\end{aligned}$$

2. Reflected Path ($n = 2$):

- Delay:

$$\tau_2 = \frac{d_2}{c} = \frac{4.5 \text{ m}}{3 \times 10^8 \text{ m/s}} = 15.0 \text{ ns}$$

- Amplitude: $a_2 = |\Gamma| \frac{d_1}{d_2} = 0.6 \times \frac{3.0}{4.5} = 0.4$.
- Reflection phase: $\phi_2 = \pi$ (from $\Gamma = -0.6$).
- Complex gain:

$$\begin{aligned}\alpha_2 &= 0.4 e^{j\pi} e^{-j2\pi(60 \times 10^9)(15 \times 10^{-9})} \\ &= 0.4 e^{j\pi} e^{-j1800\pi} = -0.4\end{aligned}$$

Delay Spread

$$\Delta\tau = \tau_2 - \tau_1 = 15.0 \text{ ns} - 10.0 \text{ ns} = 5.0 \text{ ns}$$

Step 8B: 60 GHz Case Study, Evaluating $|H(f)|$

The channel transfer function is:

$$\begin{aligned} H(f) &= \alpha_1 e^{-j2\pi f \tau_1} + \alpha_2 e^{-j2\pi f \tau_2} \\ &= 1.0 e^{-j2\pi f (10 \times 10^{-9})} - 0.4 e^{-j2\pi f (15 \times 10^{-9})} \\ &= e^{-j2\pi f \tau_1} (1.0 - 0.4 e^{-j2\pi f \Delta\tau}) \quad (\Delta\tau = 5.0 \text{ ns}) \end{aligned}$$

Magnitude Extrema:

- **Constructive** ($e^{-j2\pi f \Delta\tau} = -1$):

$$|H|_{\max} = 1.0 + 0.4 = 1.4 \implies +2.92 \text{ dB}$$

- **Destructive** ($e^{-j2\pi f \Delta\tau} = +1$):

$$|H|_{\min} = 1.0 - 0.4 = 0.6 \implies -4.44 \text{ dB}$$

Spectral Fading

- Total variation across band:

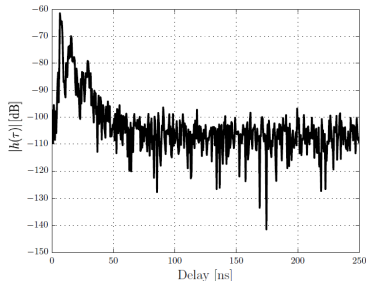
$$\Delta|H| = 2.92 - (-4.44) = 7.36 \text{ dB}$$

- Ripple frequency period:

$$\Delta f_{\text{period}} = \frac{1}{\Delta\tau} = \frac{1}{5.0 \text{ ns}} = 200 \text{ MHz}$$

- Over $B = 2 \text{ GHz}$, the transfer function executes $\frac{2000 \text{ MHz}}{200 \text{ MHz}} = 10$ fading cycles.

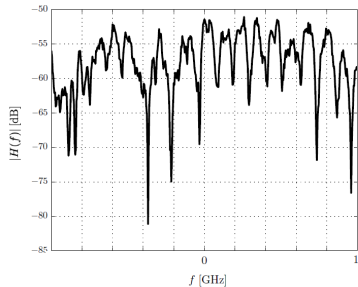
Step 9: Comparison with Experimental Data



Delay Domain: Figure 1.2 in Course

Experimental $|h(\tau)|$ (60 GHz, 2 GHz BW)

- Measured via inverse Fourier transform over 2 GHz bandwidth.
- Shows direct path at $\tau \approx 0$ followed by dense multipath trail extending to 200 ns.
- Noise floor visible at ≈ -110 dB.



Frequency Domain: Figure 1.3 in Course

Experimental $|H(f)|$ (60 GHz, 2 GHz BW)

- Plotted across baseband frequency $f \in [-1, 1]$ GHz.
- Displays severe, rapid fluctuations ranging from -52 dB down to -80 dB.
- Directly confirms destructive multi-ray cancellation across spectrum.

Step 10: Time-Variant Channels & Quasi-Static Approximation

When transmitter, receiver, or obstacles move, amplitudes and phases vary over clock-time t :

$$h(\tau, t) = \sum_{n=1}^N a_n(t) e^{j\phi_n(t)} e^{-j2\pi f_c \tau_n} \delta(\tau - \tau_n) = \sum_{n=1}^N \alpha_n(t) \delta(\tau - \tau_n)$$

The general input-output relation becomes:

$$y(t) = \int_0^\infty h(\tau, t) x(t - \tau) d\tau \quad (\text{no longer a convolution product})$$

Time-Variant Transfer Function:

$$H(f, t) = \int_{-\infty}^{\infty} h(\tau, t) e^{-j2\pi f \tau} d\tau$$

Note: $Y(f) \neq H(f, t)X(f)$ in general.

Quasi-Static Approximation

The channel is assumed **time-invariant during the transmission of each block of symbols**.

- τ (delay): fast multipath propagation ($\sim$ ns).
- t (clock time): slow macroscopic motion ($\sim$ ms).

Step 11: Engineering Impact & Multi-Carrier Mitigation

The Problem: Waveform Distortion

- Because $|H(f)|$ is not constant across bandwidth B , each spectral component $X(f)$ undergoes different complex scaling.
- In the time domain, this causes severe **Inter-Symbol Interference (ISI)**.
- Single-carrier systems require complex time-domain equalizers.

Modern Solution: Multi-Carrier (OFDM)

- In modern standards (5G NR, Wi-Fi 6/7), the wideband bandwidth B is divided into many narrow orthogonal subcarriers.
- Across each narrow subcarrier bandwidth, $|H(f)|$ is approximately constant.
- **Result:** Converts a wideband frequency-selective channel into narrowband flat-fading subchannels.

⇒ *Covered in detail in Modulation and Coding course.*

Summary: Complete Logical Chain of Derivations

- ❶ **Baseband Conversion:** Path delay τ_n and carrier f_c define the complex amplitude $\alpha_n = a_n e^{j\phi_n} e^{-j2\pi f_c \tau_n}$, sensitive to millimeter movements at 60 GHz.
- ❷ **Impulse Response:** Multipath superposition creates time dispersion: $h(\tau) = \sum_{n=1}^N \alpha_n \delta(\tau - \tau_n)$, yielding convolution $y(t) = (h * x)(t)$.
- ❸ **Fourier Duality:** Applying the Fourier transform to $y(t)$ proves $Y(f) = H(f)X(f)$ with $H(f) = \sum_{n=1}^N \alpha_n e^{-j2\pi f \tau_n}$.
- ❹ **Selectivity Proof:**
 - $\tau_n = \tau_1 \implies |H(f)| = \text{constant}$ (Flat channel).
 - $\exists \tau_n \neq \tau_m \implies |H(f)|$ fluctuates with f (Frequency-selective channel).
- ❺ **Interference Physics:** Phase difference $\Delta\Phi(f) = -2\pi f(\tau_n - \tau_m)$ rotates across bandwidth, creating fading ripples with period $\Delta f = 1/\Delta\tau$.